# Member 2 | Parts 3–4: Enhancement and performance

One fixed rule, adapted from FT5009 Lecture 8 (physical PDF pp. 16 and 21): confirm institutional Weekly Top 10 buys when the SGD total-return index is above its 126-session simple moving average (SMA), and sells when it is below. Both the current index and its SMA end on the last benchmark session strictly before the report publication date. Require 126 aligned nonmissing closes; equality or missing history rejects the slot. The total-return index avoids mechanical dividend-date price drops. Six months is a fixed medium-term smoothing horizon, not selected by maximising this sample. The lecture illustrates long/cash price-versus-SMA; we extend it symmetrically to the required long/short institutional flow slots, retaining weekly publication-driven execution. No parameter sweep or full-period optimisation is performed.

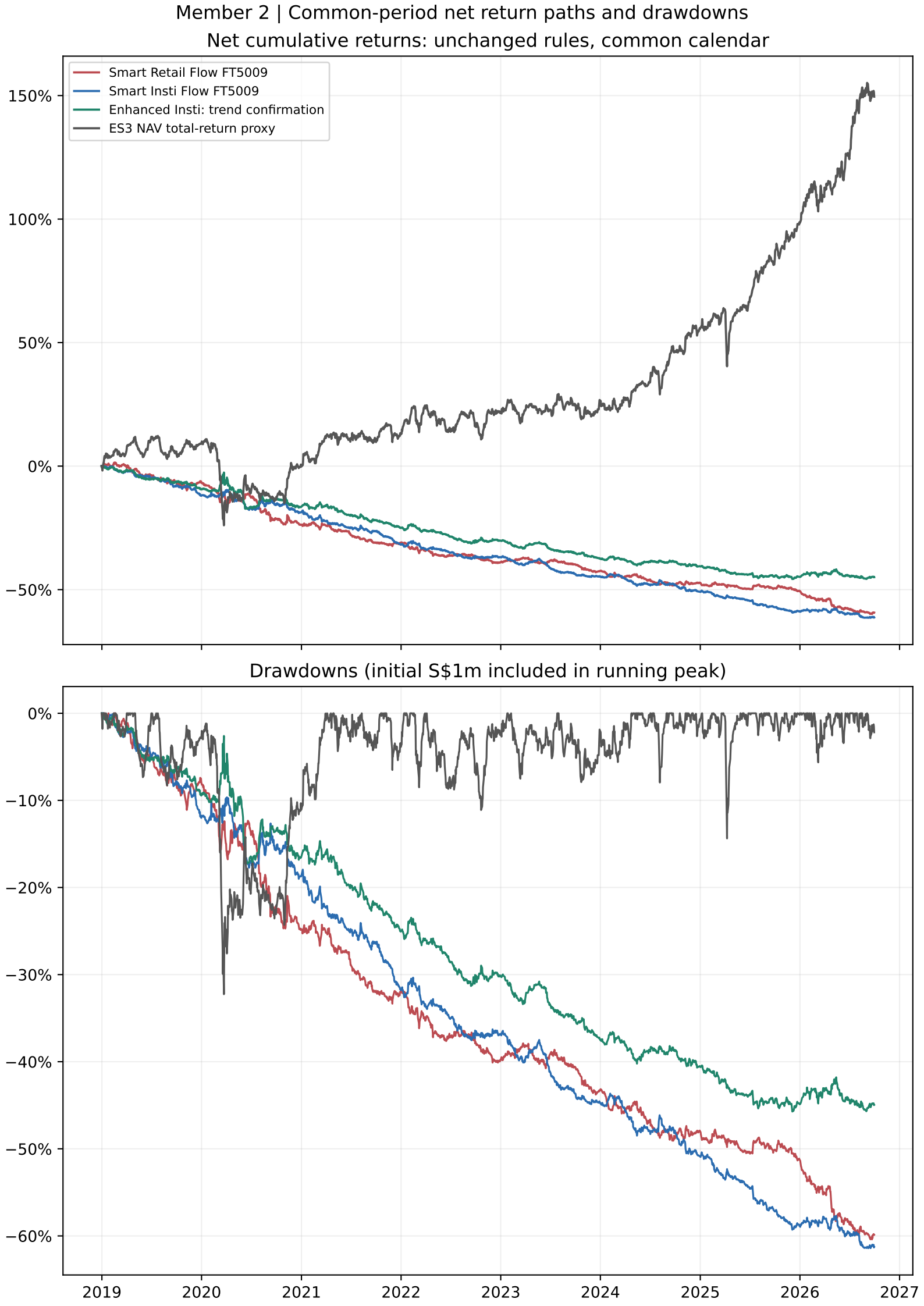
All strategies retain Member 1's same 125 fixed pre-2019 canonical histories, 396 reports and 15,840 signals; S$1m initial NAV each; 49% per side, ten 4.9% slots, 10% name cap; no redistribution of rejected/unavailable slots. Existing holdings earn execution-day returns, then rebalance at the first benchmark close strictly after publication. A rejected old holding exits at that rebalance unless zero-volume rules freeze it. Frozen holdings reserve their side budget. Cash/short proceeds earn zero, fractional total-return units and no final liquidation. All borrowable is assumed, with 3% annual calendar-day borrow on prior-close short value. No margin calls or market impact are modelled.

## Full-period NET performance (2019–September 2026)

| Strategy | Cum. | Ann. | Vol. | MDD | Sharpe | Sortino |
|---|---|---|---|---|---|---|
| Retail | -59.29% | -10.99% | 6.43% | -60.40% | -1.78 | -2.37 |
| Institutional | -61.27% | -11.56% | 5.72% | -61.41% | -2.12 | -2.74 |
| Enhanced | -44.96% | -7.45% | 5.91% | -45.72% | -1.28 | -1.70 |
| Benchmark | 149.56% | 12.58% | 13.25% | -32.26% | 0.96 | 1.36 |

On this restricted sample, retail flow has the higher full-period net cumulative return. The enhancement changes institutional net cumulative return by 16.30 percentage points (-61.27% to -44.96%). Its annualised volatility is 5.91% versus original 5.72%; maximum drawdown is -45.72%, Sharpe -1.280 and Sortino -1.701, versus original institutional Sharpe -2.119. All three strategies lose money after base costs; the enhancement reduces losses and drawdown but does not make this trading rule viable. Its mean gross exposure falls from 86.87% to 59.75%; lower exposure need not improve signal alpha.

Returns are simple daily SGD portfolio returns on a common 1,945-session calendar. Cumulative return = product(1+r)-1; annualised return = product(1+r)^(252/n)-1. Realised volatility uses sample standard deviation times sqrt(252). Maximum drawdown includes initial wealth in the peak; we report signed negative drawdowns, equivalent to the lecture's positive loss magnitude. Sharpe = mean(r)/sd(r)*sqrt(252), with zero risk-free rate. Sortino = mean(r)/sqrt(mean(min(r,0)^2))*sqrt(252), zero daily target and all n sessions in the downside denominator. Ratios with zero denominators are undefined. Ratios use arithmetic mean, not CAGR. Each subperiod compounds the existing daily path (including its first daily return); strategies are not reset or retuned at period boundaries. Subperiod drawdowns use local initial wealth; actual dollar fees retain the continuing portfolio scale.


Member 2 | Common-period net return paths and drawdowns
Net cumulative returns: unchanged rules, common calendar
Smart Retail Flow FT5009
Smart Insti Flow FT5009
Enhanced Insti: trend confirmation
ES3 NAV total-return proxy
150%
100%
50%
0%
−50%
Drawdowns (initial S$1m included in running peak)
0%
−10%
−20%
−30%
−40%
−50%
−60%
2019
2020
2021
2022
2023
2024
2025
2026
2027

# Period stability, cost stress and trading feasibility

| Net / benchmark | 2019–2021 | 2022–2023 | 2024–Sep2026 |
|---|---|---|---|
| Retail | -31.03% | -16.66% | -29.18% |
| Institutional | -31.79% | -18.79% | -30.08% |
| Enhanced | -25.04% | -16.44% | -12.13% |
| Benchmark | 12.83% | 12.36% | 96.85% |

2019–2021: institutional -31.79%, enhanced -25.04% (+6.74 percentage points). 2022–2023: institutional -18.79%, enhanced -16.44% (+2.35 percentage points). 2024–Sep2026: institutional -30.08%, enhanced -12.13% (+17.95 percentage points). The enhancement improves cumulative return in every prescribed subperiod.

Retail: gross -0.15%, net -59.29%, doubled-cost -81.72%; annual one-way turnover 51.77x NAV, mean gross exposure 85.65%. Institutional: gross -0.88%, net -61.27%, doubled-cost -83.32%; annual one-way turnover 54.01x NAV, mean gross exposure 86.87%. Enhanced: gross 2.49%, net -44.96%, doubled-cost -68.77%; annual one-way turnover 36.44x NAV, mean gross exposure 59.75%.

Retail: 785/9787 trades exceed 1% of execution-day vendor volume; maximum 31886.09%; lagged ADV21 maximum 84.97%; 25 orders exceed 100% of reported daily volume. Institutional: 543/9859 trades exceed 1% of execution-day vendor volume; maximum 6088.77%; lagged ADV21 maximum 86.92%; 18 orders exceed 100% of reported daily volume. Enhanced: 421/7415 trades exceed 1% of execution-day vendor volume; maximum 3164.38%; lagged ADV21 maximum 14.35%; 7 orders exceed 100% of reported daily volume.

Gross: zero fees/borrow. Net: 20 bps one-way transaction cost + 3% annual borrow. Stress: 40 bps + unchanged borrow. Turnover sums absolute traded notional / pre-cost NAV; annualised 252/n. All cost scenarios rerun the accounting. Liquidity is a vendor-volume proxy; actual-day volume is ex-post, ADV21 is strictly lagged. Missing slots remain cash; lower exposure is not isolated alpha.

The enhancement reduces losses in all three subperiods but remains negative after costs. The fixed sample, missing/delisted histories, candidate identifiers, vendor corporate actions, current data vintages and uncertain historical publication timestamps limit inference. ES3 official NAV total return is long-only market context, not STI total return or an executable ETF-price path. No causal investor-skill or out-of-sample claim is made. Unequal realised exposure matters. Volume uses vendor price/volume pairs: orders above 100% of daily volume cannot be fully filled within that day on the recorded basis. This requires execution constraints or vendor investigation; the current paths are unconstrained accounting simulations. Lagged ADV is a proxy, not guaranteed liquidity. Assumed universal borrow, zero cash interest, no recalls, margin or nonlinear impact further limit realism. All three net strategies and doubled-cost paths lose money: none establishes trading viability.

Sources: FT5009 AY2026/27 assignment; SGX Weekly Top 10 and Member 1 collected data/audit; FT5009 L08 physical PDF pp. 16/21 (SMA rule), 42–47 (performance), 52–54 (pitfalls); L02 pp. 16–18 (data/PIT); L07 pp. 12/16–17 (impact/depth). AI: OpenAI Codex assisted with code, analysis and text. Full formulas, tables and risk inputs accompany this chapter.